THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 25	Intercepts are c and $-c/m$; area $= \frac{1}{2}\lvert c\rvert \cdot \lvert c/m\rvert$.
2	**(a)** 4	Equate the two expressions.
3	**(a)** 25	Evaluate the inner value first.
4	**(a)** 12	Equate $a_1 + b_1x = a_2 + b_2x$.
5	**(a)** 7	$kz - c = 0$.
6	**(c)** $b = 0$	Reason about linear polynomials.
7	**(d)** 64	Find each term.
8	**(d)** 17	Solve the linear equation.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$-\frac{-12}{3} = 4$; R is the general rule that gives it.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Putting $y = 0$ gives $x = 3$, so A is true; putting $x = 0$ gives $y = -6$, so R is true. R gives the other intercept, so it does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$200 + 30x = 440 + 10x$ (1); $x = 12$ units (1).	1+1
12	$p(p(x)) = 3(3x + 4) + (4) = 9x + 16$ (1); Value $= 34$ (1).	1+1
13	Intercepts: $y = 14$ and $x = -7$ (1); Area $= \frac{1}{2} \times 14 \times 7 = 49$ sq. units (1).	1+1
14	$150 + 25x = 390 + 5x$ (1); $x = 12$ units (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$5x + 2 = 2x + 11$ (1); $x = 3$ (1); $y = 17$ (1).	1+1+1
16	$p(3) - p(1) = 2a = 8$, so $a = 4$; $b = 8$, $p(x) = 4x + 8$ (1); Zero $x = -2$ (1); $p(6) = 32$ (1).	1+1+1
17	$14n + 3 < 143$ gives $n < 10$ (1); Terms: $n = 1, 2, \ldots, 9$, number $= 9$ (1); $a_1 + a_2 = 17 + 31 = 48$ (1).	1+1+1
	OR: $7x - 3 = 3x + 9$ (1); $x = 3$ (1); $y = 18$ (1).	1+1+1
18	$p(3) - p(1) = 2a = 10$, so $a = 5$; $b = -15$, $p(x) = 5x - 15$ (1); Zero $x = 3$ (1); $p(4) = 5$ (1).	1+1+1
	OR: $11n - 4 < 128$ gives $n < 12$ (1); Terms: $n = 1, 2, \ldots, 11$, number $= 11$ (1); $a_1 + a_2 = 7 + 18 = 25$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $C(x) = 12x + 50$ (1). (ii) $C(10) = 120 + 50 =$ Rs 170 (1). (iii) $12x + 50 = 230 \Rightarrow x = 15$ km (1). (iv) Slope $= 12$: the rate of Rs 12 per km (1); y-intercept $= 50$: the fixed charge of Rs 50 (1).	1+1+1+2
20	$A(x) = 12x + 30$, $B(x) = 7x + 90$ (1); $12x + 30 = 7x + 90$ gives $x = 12$ km (1); Common fare $= 174$ (1); For 8 km: A $= 126$, B $= 146$ (1); A is cheaper (1).	1+1+1+1+1
	OR: Slope $= 5$ (1); Equation $y = 5x - 10$ (1); y-intercept -10, x-intercept 2 (1); Area $= \frac{1}{2} \times 10 \times 2$ (1); $= 10$ sq. units (1).	1+1+1+1+1

Mistakes to avoid:

- Writing the zero of $ax + b$ as $\frac{b}{a}$ instead of $-\frac{b}{a}$.
- Taking the slope of $y = ax + b$ to be b; the slope is the coefficient a and b is the y-intercept.
- Calling $x^2 - 5$ or $\frac{1}{x} + 2$ a linear polynomial; the degree must be exactly 1.
- Finding the x-intercept by putting $x = 0$ (that gives the y-intercept); for the x-intercept put $y = 0$.
- In the nth term of a sequence, writing $t_n = dn + t_1$ instead of $dn + (t_1 - d)$, so that t_1 does not match.
- Reading the fixed charge as the slope, and the per-unit rate as the intercept, in word problems.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 14	Evaluate the inner value first.
2	**(d)** 12	Equate $a_1 + b_1x = a_2 + b_2x$.
3	**(a)** 8	$kz - c = 0$.
4	**(a)** a horizontal line	Reason about linear polynomials.
5	**(b)** 69	Find each term.
6	**(c)** 22	Solve the linear equation.
7	**(d)** 48	Intercepts are c and $-c/m$; area $= \frac{1}{2}\lvert c\rvert \cdot \lvert c/m\rvert$.
8	**(b)** 3	Equate the two expressions.
9	**(c)** A is true but R is false.	The slope is the coefficient of x, i.e. -4, so A is true; R is false because b is the y-intercept.
10	**(d)** A is false but R is true.	$x^2 - 5$ has degree 2, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$p(p(x)) = 2(2x - 7) + (-7) = 4x - 21$ (1); Value $= -1$ (1).	1+1
12	Intercepts: $y = 20$ and $x = -5$ (1); Area $= \frac{1}{2} \times 20 \times 5 = 50$ sq. units (1).	1+1
13	$90 + 40x = 330 + 20x$ (1); $x = 12$ units (1).	1+1
14	$p(p(x)) = 4(4x + 3) + (3) = 16x + 15$ (1); Value $= 31$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$4x + 8 = -2x - 4$ (1); $x = -2$ (1); $y = 0$ (1).	1+1+1
16	$p(3) - p(1) = 2a = 12$, so $a = 6$; $b = 12$, $p(x) = 6x + 12$ (1); Zero $x = -2$ (1); $p(3) = 30$ (1).	1+1+1
17	$16n + 5 < 149$ gives $n < 9$ (1); Terms: $n = 1, 2, \ldots, 8$, number $= 8$ (1); $a_1 + a_2 = 21 + 37 = 58$ (1).	1+1+1
	OR: $6x + 1 = x + 21$ (1); $x = 4$ (1); $y = 25$ (1).	1+1+1
18	$p(3) - p(1) = 2a = 6$, so $a = 3$; $b = -21$, $p(x) = 3x - 21$ (1); Zero $x = 7$ (1); $p(10) = 9$ (1).	1+1+1
	OR: $12n - 2 < 178$ gives $n < 15$ (1); Terms: $n = 1, 2, \ldots, 14$, number $= 14$ (1); $a_1 + a_2 = 10 + 22 = 32$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $12 - 7 = 5$ (1). (ii) $t_n = 5n + 2$ (1). (iii) $5n + 2 = 87 \Rightarrow n = 17$ (1). (iv) $5n + 2 = 100 \Rightarrow n = \frac{98}{5}$, not a whole number (1), so no week gives exactly Rs 100 (1).	1+1+1+2
20	$A(x) = 9x + 20$, $B(x) = 4x + 80$ (1); $9x + 20 = 4x + 80$ gives $x = 12$ km (1); Common fare $= 128$ (1); For 6 km: A $= 74$, B $= 104$ (1); A is cheaper (1).	1+1+1+1+1
	OR: Slope $= 6$ (1); Equation $y = 6x - 12$ (1); y-intercept -12, x-intercept 2 (1); Area $= \frac{1}{2} \times 12 \times 2$ (1); $= 12$ sq. units (1).	1+1+1+1+1

Mistakes to avoid:

- Writing the zero of $ax + b$ as $\frac{b}{a}$ instead of $-\frac{b}{a}$.
- Taking the slope of $y = ax + b$ to be b; the slope is the coefficient a and b is the y-intercept.
- Calling $x^2 - 5$ or $\frac{1}{x} + 2$ a linear polynomial; the degree must be exactly 1.
- Finding the x-intercept by putting $x = 0$ (that gives the y-intercept); for the x-intercept put $y = 0$.
- In the nth term of a sequence, writing $t_n = dn + t_1$ instead of $dn + (t_1 - d)$, so that t_1 does not match.
- Reading the fixed charge as the slope, and the per-unit rate as the intercept, in word problems.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 11	$kz - c = 0$.
2	**(d)** $2x + 6$	Reason about linear polynomials.
3	**(a)** 73	Find each term.
4	**(a)** 19	Solve the linear equation.
5	**(b)** 45	Intercepts are c and $-c/m$; area $= \frac{1}{2}\lvert c\rvert \cdot \lvert c/m\rvert$.
6	**(b)** 6	Equate the two expressions.
7	**(a)** 9	Evaluate the inner value first.
8	**(a)** 15	Equate $a_1 + b_1x = a_2 + b_2x$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$t_n = 3n + 2$ gives $5, 8, 11, 14$; the constant difference 3 is the coefficient of n.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$p(2) = 10 - 2 = 8$, so A is true; R is true but is not the reason for the value.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Intercepts: $y = 12$ and $x = -2$ (1); Area $= \frac{1}{2} \times 12 \times 2 = 12$ sq. units (1).	1+1
12	$500 + 55x = 740 + 35x$ (1); $x = 12$ units (1).	1+1
13	$p(p(x)) = 5(5x - 2) + (-2) = 25x - 12$ (1); Value $= 63$ (1).	1+1
14	Intercepts: $y = 50$ and $x = -5$ (1); Area $= \frac{1}{2} \times 50 \times 5 = 125$ sq. units (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$9x + 4 = 4x + 34$ (1); $x = 6$ (1); $y = 58$ (1).	1+1+1
16	$p(3) - p(1) = 2a = 14$, so $a = 7$; $b = 14$, $p(x) = 7x + 14$ (1); Zero $x = -2$ (1); $p(2) = 28$ (1).	1+1+1
17	$18n + 1 < 145$ gives $n < 8$ (1); Terms: $n = 1, 2, \ldots, 7$, number $= 7$ (1); $a_1 + a_2 = 19 + 37 = 56$ (1).	1+1+1
	OR: $8x - 5 = 3x + 15$ (1); $x = 4$ (1); $y = 27$ (1).	1+1+1
18	$p(3) - p(1) = 2a = 16$, so $a = 8$; $b = -16$, $p(x) = 8x - 16$ (1); Zero $x = 2$ (1); $p(5) = 24$ (1).	1+1+1
	OR: $13n + 6 < 149$ gives $n < 11$ (1); Terms: $n = 1, 2, \ldots, 10$, number $= 10$ (1); $a_1 + a_2 = 19 + 32 = 51$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $h(0) = 120$ cm (1). (ii) $h(10) = 120 - 40 = 80$ cm (1). (iii) $120 - 4t = 0 \Rightarrow t = 30$ minutes (1). (iv) Slope $= -4$; intercepts $(0, 120)$ and $(30, 0)$ (1); the height falls by 4 cm every minute (1).	1+1+1+2
20	$A(x) = 13x + 35$, $B(x) = 8x + 95$ (1); $13x + 35 = 8x + 95$ gives $x = 12$ km (1); Common fare $= 191$ (1); For 10 km: A $= 165$, B $= 175$ (1); A is cheaper (1).	1+1+1+1+1
	OR: Slope $= 7$ (1); Equation $y = 7x + 21$ (1); y-intercept 21, x-intercept -3 (1); Area $= \frac{1}{2} \times 21 \times 3$ (1); $= \frac{63}{2}$ sq. units (1).	1+1+1+1+1

Mistakes to avoid:

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- Calling $x^2 - 5$ or $\frac{1}{x} + 2$ a linear polynomial; the degree must be exactly 1.
- Finding the x-intercept by putting $x = 0$ (that gives the y-intercept); for the x-intercept put $y = 0$.
- In the nth term of a sequence, writing $t_n = dn + t_1$ instead of $dn + (t_1 - d)$, so that t_1 does not match.
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